

ARTIFICIAL INTELLIGENCE

Lecture Set 01
Introduction and History of AI

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Course Contents

→ 1. Introduction to AI

2. Agents and Environments
3. Problem Solving by Searching
4. Uninformed Search
5. Informed Search
6. Local Search
7. Constraint Satisfaction Problems
8. Adversarial Search
9. Logical Agents
10. First-order Logic
11. Uncertain Knowledge and Reasoning
12. Supervised, Unsupervised, and Reinforcement Learning
13. Recent Trends in AI and Applications of AI Algorithms

Course Description

- This course introduces students to the basic knowledge representation, problem solving, and learning methods of artificial intelligence.
- Upon completion, students should be able to develop intelligent systems by assembling solutions to concrete computational problems; understand the role of knowledge representation, problem solving, and learning in intelligent-system engineering; and appreciate the role of problem solving, vision, and language in understanding human intelligence from a computational perspective.

Grading Policy and Marks Distribution

- Absolute grading

Assessment item	Number	Weightage
Sessional	2	15+15
Project	1	10
Assignments	4	10
Final exam	1	50

Recommended book

1. Stuart Russell and Peter Norvig, Artificial Intelligence. A Modern Approach, 3rd edition, Prentice Hall, Inc., 2010.

Table of Contents

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- Foundations of AI
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- Concept of neurons and neural networks
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- AI in industry
- State of the Art

What is Intelligence?

- In the popular sense, intelligence is defined as the **ability to learn** and **apply knowledge** to manipulate your environment, as well as the **ability to reason** and have abstract thought.
- Other definitions include:
 - *Adaptability to a new environment*
 - *The ability to evaluate and judge*
 - *The ability to comprehend complex ideas*
 - *The capacity for original and productive thought*
 - *The ability to learn quickly and learn from experience.*



What is Artificial Intelligence?

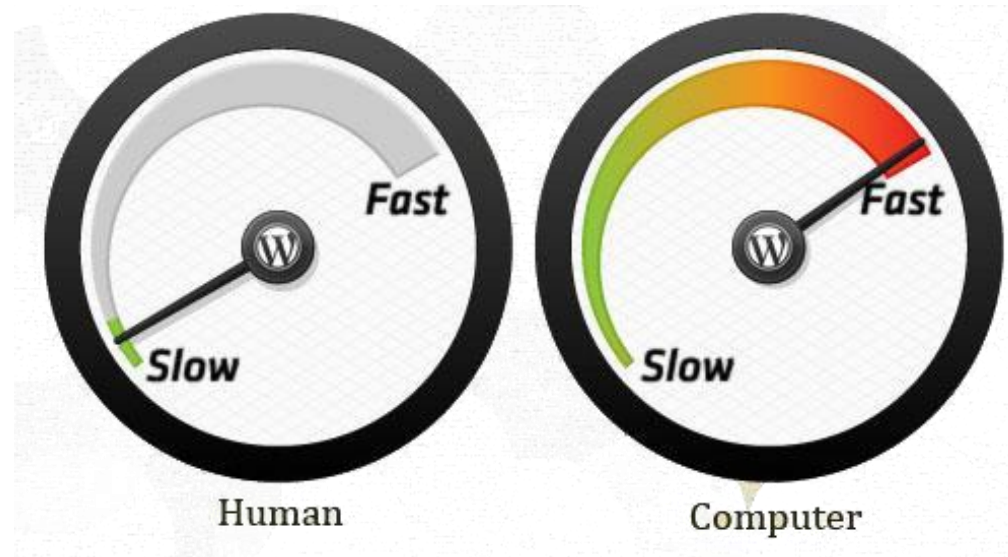
- Artificial Intelligence is the study of how to make computers do things at which, at the moment, people are better.
- Embedding human intelligence in to the machine so that they can think like humans, behave like humans and make decisions like humans.



Tasks: Machines better than humans

1. Speed

- Computer can perform any calculation in split of a second.
- Human consumes more time to do the same task.



Tasks: Machines better than humans

2. Accuracy

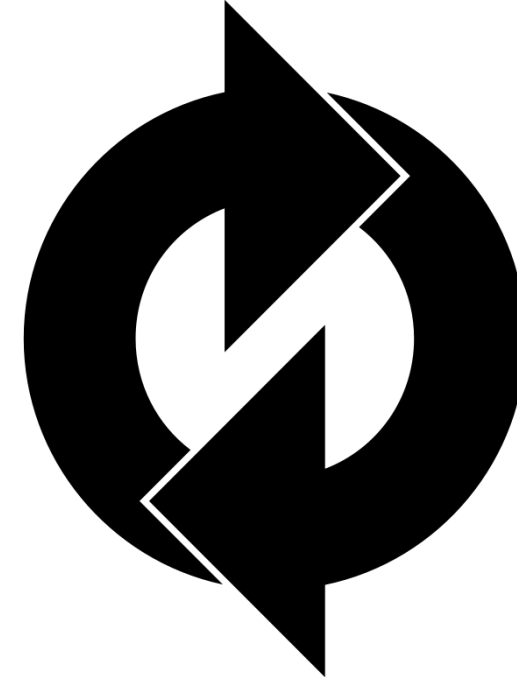
- Computer are very much accurate unless any software or hardware failure occurs.
- Computer accurately performs what it is told to do.
- Human is prone to errors.



Tasks: Machines better than humans

3. Delegacy

- Computer can perform specific task repetitively infinite number of times with uniform accuracy.
- Human gets bored with repetitive tasks.



Tasks: Machines better than humans

4. Run for a long period of time

- Computer can run for a very long period of time (once started).
- It does not need any rest.
- Human needs some rest and energy to do a task.



Tasks: Machines better than humans

5. Multitasking

- Computer can perform several different tasks at a time at very high speed and tremendous accuracy.
- Human can not concentrate mind on multiple tasks.



Other definitions of AI

Thinking humanly	Thinking rationally
“The exciting new effort to make computers think... machines with minds, in the full and literal sense.” <i>(Haugeland, 1985)</i>	“The study of mental faculties through the use of computational models.” <i>(Charniak & McDermott, 1985)</i>
Acting humanly	Acting rationally
“The study of how to make computers do things which, at the moment, people are better.” <i>(Rich & Knight, 1991)</i>	“Computational Intelligence is the study of the design of intelligent agents.” <i>(Poole et al., 1998)</i>

Foundations of AI

- Many disciplines contribute to the goal of creating/modelling intelligent entities:
 - *Philosophy (nature of belief, rationality, etc)*
 - Where does knowledge come from?
 - How does knowledge lead to action?
 - *Mathematics*
 - What are the formal rules to draw valid conclusions?
 - What can be computed?
 - *Economics*
 - How should we make decisions so as to maximize payoff?

Foundations of AI Cont'd

- Many disciplines contribute to the goal of creating/modelling intelligent entities:
 - *Neuroscience*: How do brains process information?
 - *Psychology (human reasoning)*: How do humans and animals think and act?
 - *Computer engineering*: How can we build an efficient computer?
 - *Control theory and cybernetics*: How can artifacts operate under their own control?
 - *Linguistics*: How does language relate to thought?

Goals of AI

- Two main goals of AI:

- *To understand human intelligence better. We test theories of human intelligence by writing programs which emulate it.*
- *To create useful “smart” programs able to do tasks that would normally require a human expert.*

Application areas of AI

- Game Playing
- Automated reasoning and theorem proving
- Expert Systems
- Natural Language Understanding
- Modelling Human Performance
- Planning and Robotics
- Machine Learning
- Computer Vision
- Control Systems
- Optimization

History of AI

1. **1950: Turing's Influence**

Alan Turing introduces key concepts in AI with "Computing Machinery and Intelligence."

2. **1956: Birth of AI**

Dartmouth meeting officially adopts the term "Artificial Intelligence."

3. **1950s: Initial Promise**

Early AI programs emerge, showcasing promise.

Samuel's checkers program and Newell & Simon's Logic Theory contribute to the field.

4. **AI Winter (1970s-1980s):**

Funding and interest decline due to overpromised results and lack of significant progress.

5. **1986: Rise of Machine Learning**

Neural networks regain popularity.

Significant advances in machine learning algorithms and applications mark this period.

6. **1990: Role of Uncertainty**

Bayesian networks gain prominence as a knowledge representation framework.

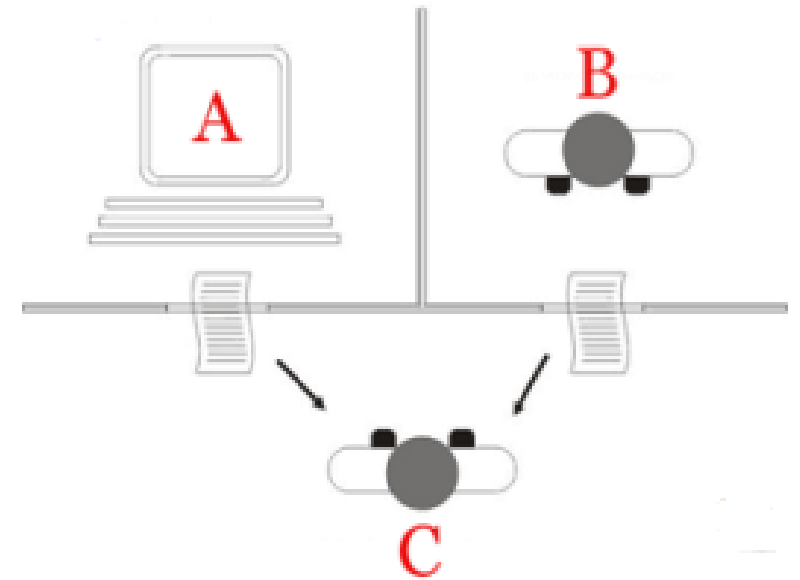
7. **1995: AI as Science**

Integration of learning, reasoning, and knowledge representation.

AI methods find applications in diverse domains, including vision, language, and data mining.

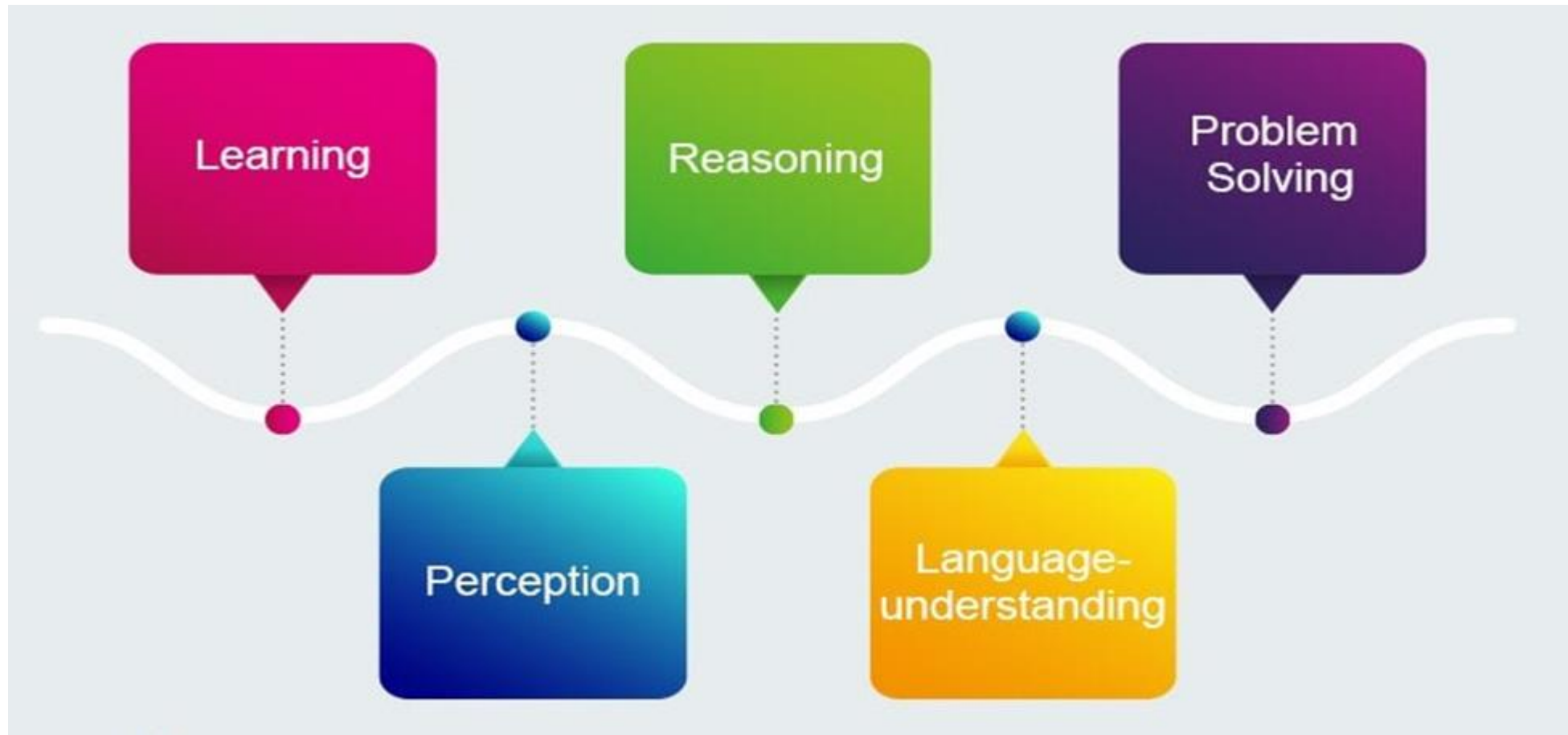
Turing Test

- 3 rooms contain: a person, a computer, and an interrogator.
- The interrogator can communicate with the other 2 by teletype (to avoid the machine imitate the appearance or voice of the person).
- The interrogator tries to determine which is the person and which is the machine.
- The machine tries to fool the interrogator to believe that it is the human, and the person also tries to convince the interrogator that it is the human.
- If the machine succeeds in fooling the interrogator, then conclude that the machine is intelligent.
- Goal is to develop systems that are human-like.

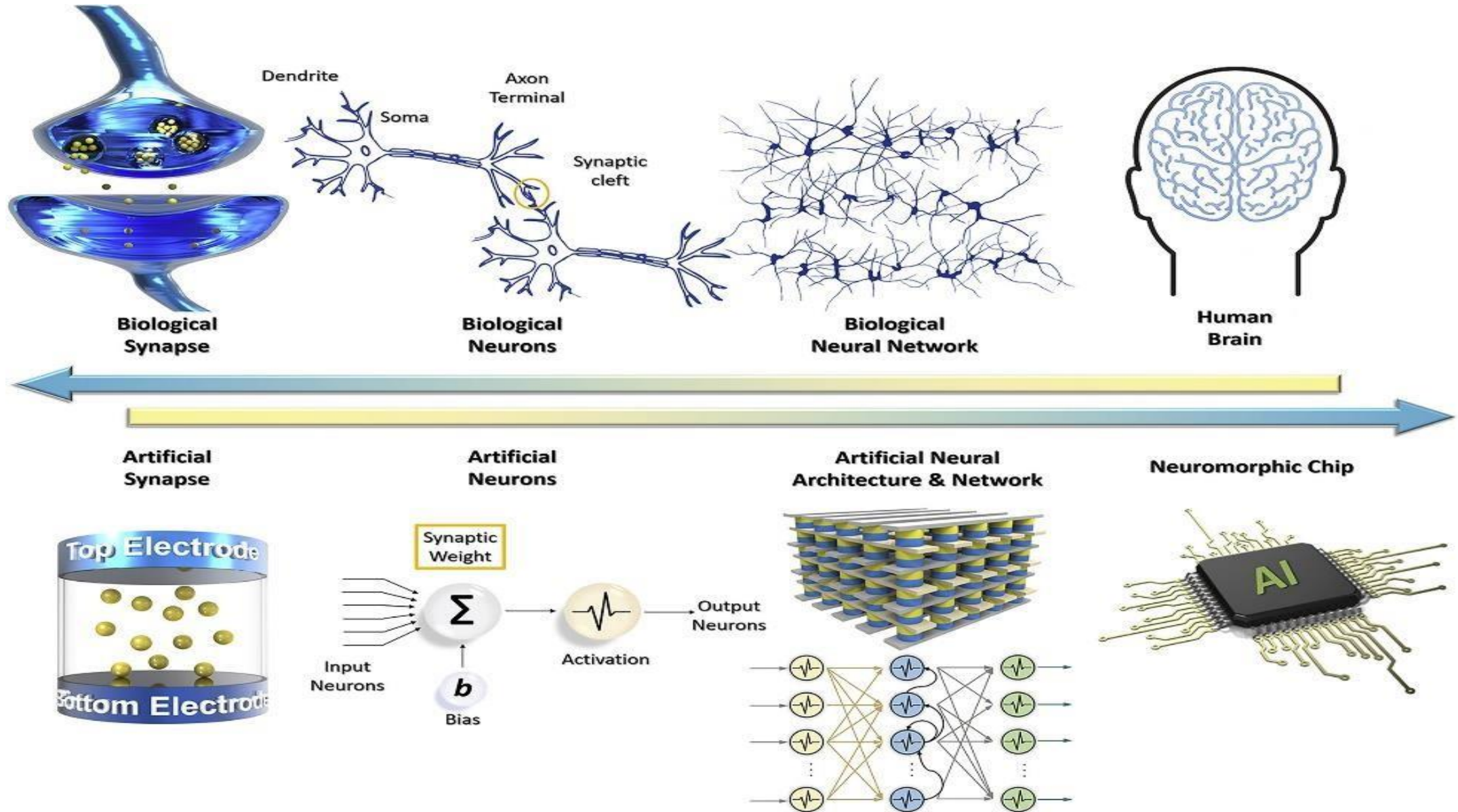


Basic Components of AI

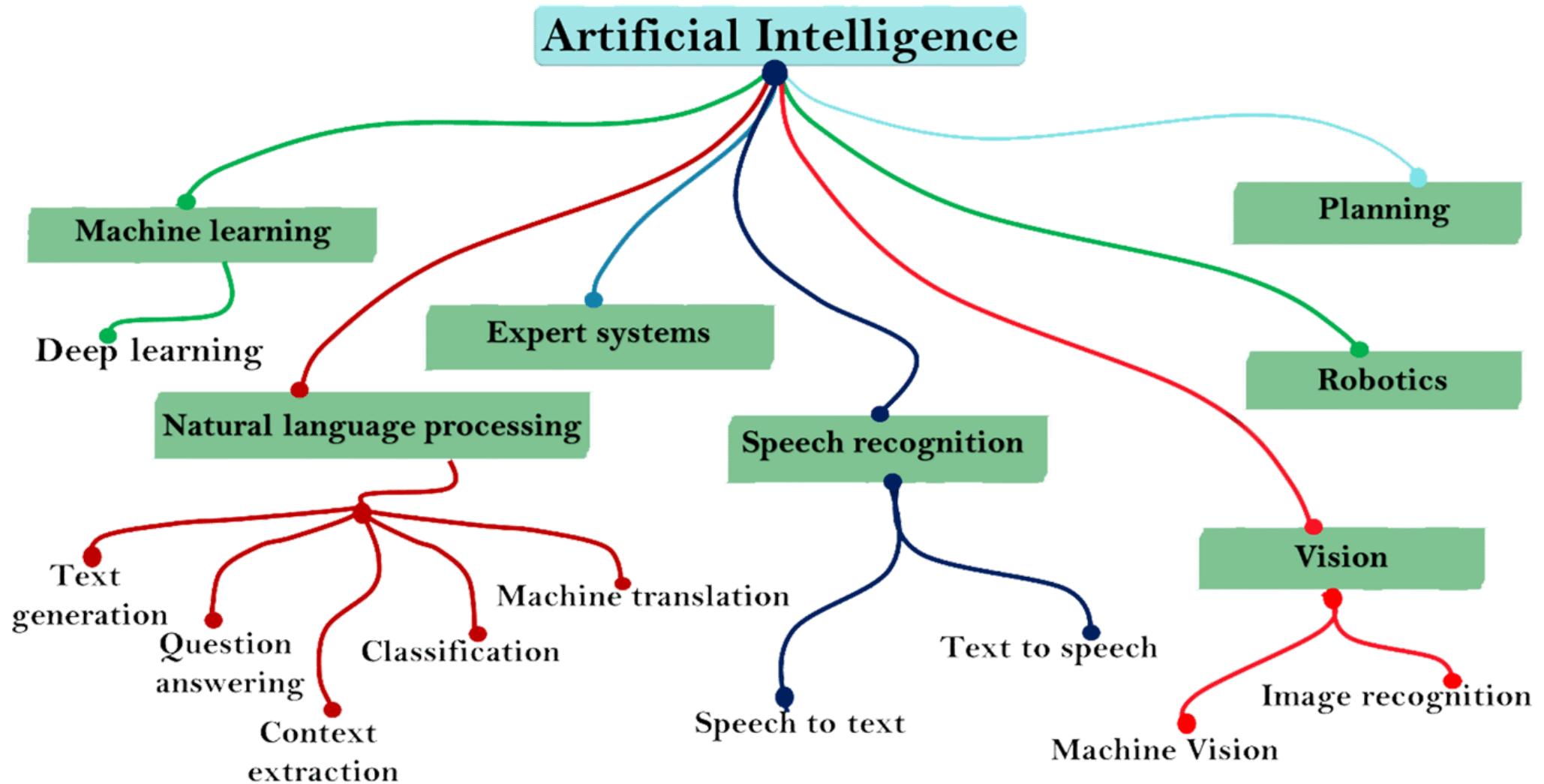
The basic components of AI refer to the fundamental elements or building blocks that constitute artificial intelligence systems.



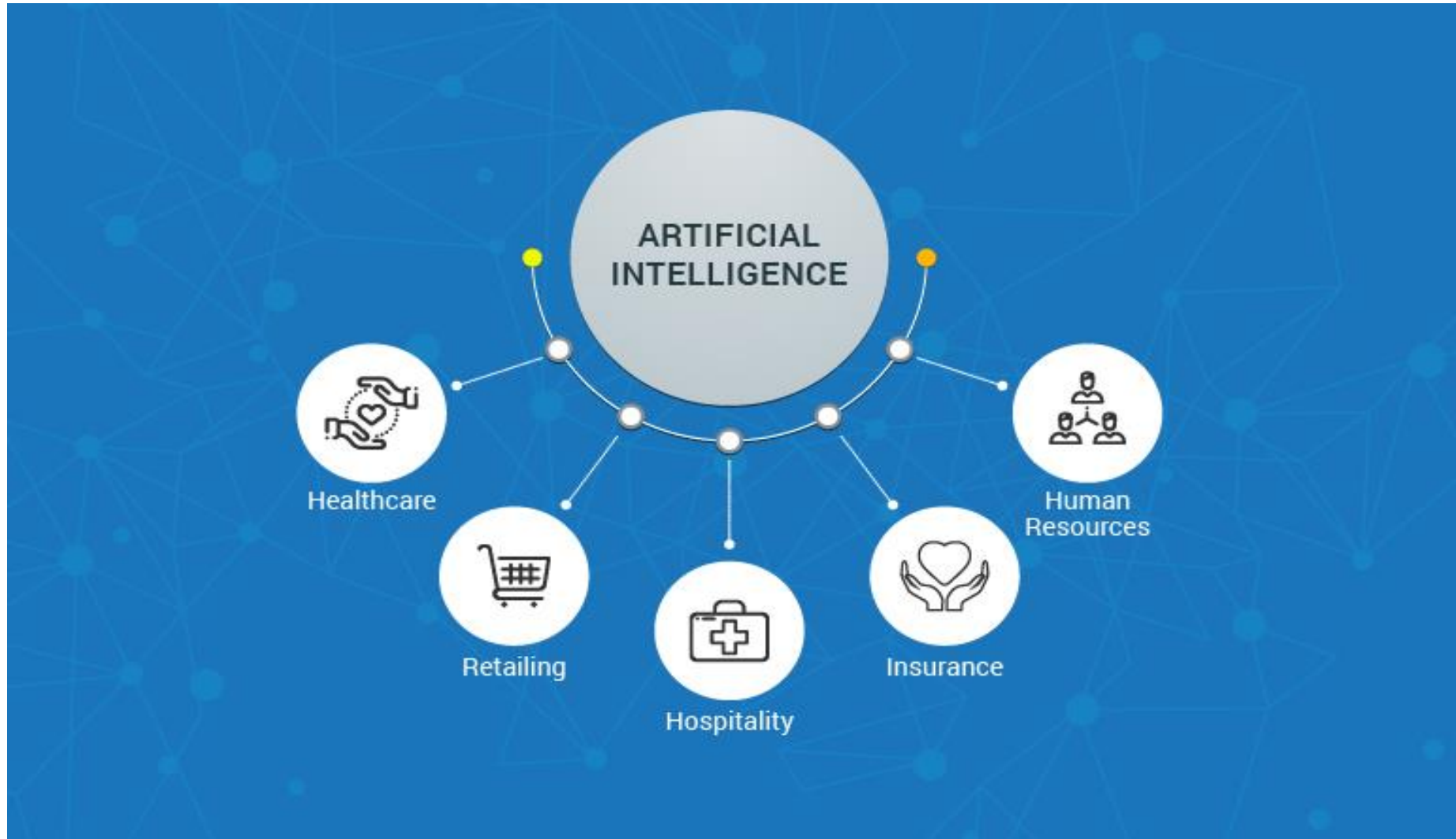
Concept of Neurons and Neural Networks



Branches of AI



AI in Industry



State of the Art

- Robotic vehicles
- Speech recognition
- Autonomous planning and scheduling
- Game playing
- Spam fighting
- Logistics planning
- Robotics
- Machine translation

Questions

1. What is artificial intelligence (AI), and how does it differ from traditional computer programming?
2. Why is AI considered a transformative technology, and what are some of its potential applications?
3. What are the key principles underlying AI, and how do they contribute to its development?
4. Discuss the importance of problem-solving, knowledge representation, and reasoning in AI systems.
5. Trace the historical development of AI from its early beginnings to the present day.
6. What were some significant milestones in the history of AI, and how did they shape the field?
7. Explain the fundamental components of an AI system, including perception, reasoning, and action.
8. How do data, algorithms, and computational power interact to enable AI applications?
9. What are neurons, and how do they function in biological brains?
10. Describe the architecture and functioning of artificial neural networks (ANNs) and their role in AI.
11. Discuss the main branches of AI, such as machine learning, natural language processing, computer vision, and robotics.
12. How do these branches complement each other, and what are some interdisciplinary applications?
13. How is AI being used in various industries, such as healthcare, finance, manufacturing, and transportation?
14. What are some benefits and challenges associated with the adoption of AI technologies in industry?
15. What are some recent advancements in AI research and development?
16. Discuss cutting-edge AI techniques and technologies, such as deep learning, reinforcement learning, and generative adversarial networks (GANs).